



# DIVIJ SINGLA

Third Year Undergraduate  
Electrical Engineering  
Indian Institute of Technology Kanpur

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| Degree     | Institute                               | Year      | CPI/%    |
|------------|-----------------------------------------|-----------|----------|
| Graduation | Indian Institute of Technology Kanpur   | 2021-2025 | 8.3/10.0 |
| CBSE XII   | Modern Vidya Niketan, Sec-17, Faridabad | 2021      | 97%      |
| CBSE X     | Modern Vidya Niketan, Sec-17, Faridabad | 2019      | 98.6 %   |

## ACADEMIC ACHIEVEMENTS

- **All India Rank 723**, JEE Advanced 2021 among 0.2 Million shortlisted candidates
- **All India Rank 189**, JEE Main 2021 among 1.2 Million candidates
- Recipient of the prestigious **KVPY Fellowship 2021** by IISc, Bangalore, secured **All India Rank 650**
- Awarded the **CBSE Certificate of Merit**, for being among **top 0.1% students all over India** in Science and Mathematics

## PROJECTS

### Block by Block

Dec'22

Self Project

- Constructed a **Blockchain** from scratch on JS with all of the key parameters like **Timestamp**, **Nonce**, **Difficulty**, etc. inspired from the concepts of Bitcoin's blockchain.
- Achieved stable avg **mining time of 2 sec** by implementing a **dynamic difficulty** adjusting mechanism for nonce discovery
- Developed a **fork conflict resolution** system based on the **longest chain rule** to determine the chain with the most accumulated **Proof of Work**, effectively managing situations where multiple miners mine a block simultaneously

### Spacings in Randomised Data

Jan '24 - Apr '24

Course Project, Randomised Algorithms | Prof. Surender Baswana

- Determined **expected length** of smallest interval for random, uniform, independent point placement on a unit length segment
- Derived the result by rigorous mathematical proof using the principles of expected value, **continuous distributions**, etc.
- Established relations between distribution of interval lengths with **exponential distribution** to find new results & insights
- Complemented theoretical findings with **empirical** validation via computer simulations to ensure accuracy of analytic results

### Attacking Cryptosystems

Jan '23 - May'23

Course Project, Modern Cryptology | Prof. Manindra Agarwal

- **Decrypted many Ciphertexts** based on complex cryptosystems as a part of an SSH Text-Adventure Game
- Explored and gained proficiency in Private and Public Key Encryption Algorithms, like **DES**, **AES**, **RSA**, and **SHA**
- Investigated **Key Exchange Mechanisms**, Post-Quantum Cryptography, Digital Signatures, and **ZK Proofs**

### DeepDive into Web3

May '22- Jul '22

Science and Technology Council, IIT Kanpur

- Researched concepts like Proof of Work, Mining, Nonce, **Gas**, **Smart Contracts**, Blockchains, Web3, **Metamask**, **NFTs**
- Deployed Smart-Contracts on **Ethereum Testnet**, used libraries like hashlib and web3.js, ethers.js in Python and JS
- Implemented a **Decentralised Bank dApp** to give loans on a credit-based system from a pool of money deposited by lenders

### Ripple Room

Dec'22

Self Project

- Implemented a **Voice Chatroom** Application using Javascript in **NodeJS** runtime environment, Socket.IO, and HTML/CSS
- Interactive interface pings users when they receive a message, displays the online users, and also **supports text messages**
- **PaySettle**: Working to expand the application functionality through the integration of rev.ai to make **cryptocurrency payments** which can be securely registered on the blockchain created by "Block-by-Block" project above

## ENTREPRENEURIAL EXPERIENCE

### FoodKnights

Built and Run a startup

Mar'23-May'23

- Identified **broken Supply-Chain** between Campus Residents and Food Delivery Companies like Zomato and Swiggy due to a sudden change in campus regulations which didn't allow delivery personnel inside the campus after 11 PM
- Successfully reverse-engineered the **API calls** of Swiggy's live website to obtain up-to-date restaurant and dish information
- Developed a web app that **dynamically updates** restaurant menus and availability using the acquired **real time data**
- Programmed a **Python bot** to automate order placement on Swiggy, with delivery address set as the Main Gate of Campus
- The remaining delivery process from the Main Gate to the Resident's Hostel was coordinated by a team of 2 IITK students
- Deployed the website on **Google Cloud Platform (GCP)** with backend framework **Flask** and **PostgreSQL** database

## WORK EXPERIENCE

### Atlassian India LLP

Software Engineer Intern

Ongoing

- Actively engaging in hands-on learning experience with seniors to develop robust backend applications
- Gaining proficiency in Spring Boot, AWS, etc.

MENTORSHIP EXPERIENCE

Project Mentor

Exploiting Blockchain with openAI APIs

ACA, IITK  
May'23 - July'23

- Led the development of an **OpenAI NFT Marketplace** on Solana Devnet enabling minting and listing of AI-infused NFTs
- Minted NFTs utilizing **Dall-E** generated images as foundational digital assets, leveraging the unique outputs of the model
- **Mentored** a team of more than **20 IIT Kanpur undergraduate** students of first and second year
- Facilitated student learning by designing sub-tasks like **Block Mining** and **Avatar Generation**, laying a solid foundation

POSITION OF RESPONSIBILITY

Marketing Organiser, Antaragni'23

Antaragni : Annual Cultural Festival of IIT Kanpur

- Secured **multiple** sponsorship deals for Antaragni'23 and effectively coordinated with brands before, during and after fest
- **Negotiated** sponsorship agreements with brands, optimizing terms and conditions to maximize benefits for both the parties
- Co-led a dynamic team of over 60 individuals, fostering **collaboration** and ensuring smooth execution of marketing strategies

SKILLS

- **Languages** : C, C++, Python, Javascript
- **Libraries** : NodeJS, crypto-js, Selenium, hashlib, web3.js, ethers.js, bs4
- **Frameworks** : ExpressJS, Flask
- **Utilities** : Git, Github, HTML, CSS, Socket.IO, rev.ai, Jinja2

MISCELLANEOUS

- Achieved **multiple medals** in various SOF Olympiads and consistently advanced to Stage-2 qualifications
- Certificate of Meritorious Performance in International **GK** Qualifier (NOF), **School Rank 1**, State Rank 48
- Achieved **City Rank 1**, International Rank 297 in 14th NCO (National Cyber Olympiad) organized by SOF
- Represented my school in District Level Open Roller **Skating Championship** (Junior)
- Participated in intra-school swimming competitions
- **Guitarist** for School Band in Itihas Anveshan 2017 Nationals team, **Runners-Up** in **Song of the Year Category** of the prestigious National Music Competition
- Achieved **first place** and received a memento in the Inter-House **Debate** Competition hosted by the school.
- Achieved **first place** and received a memento in the Inter-House **Quiz** Competition hosted by the school.
- Attended a Model United Nations Conference (**MRMUN 2017**) and represented Bolivia as their delegate in the Committee of Narcotics and Drugs
- Received **Scholar** recognition at school, awarded **distinguished purple blazer pocket** for academic excellence for three consecutive years

RELEVANT COURSES

o ONGOING, \* EXCELLENT PERFORMANCE

|                            |                                               |                                         |                           |
|----------------------------|-----------------------------------------------|-----------------------------------------|---------------------------|
| Probability and Statistics | Data Structures & Algorithms*                 | Randomised Algorithms                   | Parallel Computing        |
| Electrodynamics*           | Laboratory Physics*                           | Intro to Physics*                       | Intro to Electrical Engg* |
| Chemistry Laboratory*      | Introduction to Machine Learning <sup>o</sup> | Partial Differential Equations*         | Modern Cryptology         |
| Microelectronics           | Intro to Biology*                             | Chemistry Laboratory*                   | Psychology and Cognition  |
| Fundamentals of Computing  | Signals, Systems and Networks                 | Linear Algebra & Differential Equations |                           |